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Overview of analogue science activities at the McGill Arctic Research Station, Axel Heiberg Island, Canadian High Arctic

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ABSTRACT

The Canadian High Arctic contains several of the highest fidelity Mars analogue sites in the world. Situated at nearly 80° north, Expedition Fjord on Axel Heiberg Island is located within a polar desert climate, with the surrounding landscape and conditions providing an invaluable opportunity to examine terrestrial processes in a cold, dry environment. Through the Canadian Space Agency's Analogue Research Network program, scientific activities based out of the McGill Arctic Research Station (M.A.R.S.) are extremely broad in scope, representing physical, biological, and technological investigations. Some of the most unique hydrogeologic features under investigation near M.A.R.S. are a series of cold saline springs that maintain liquid-state flow year round regardless of air temperature. Previous studies have examined their geomorphic relation to discharge-related formations, water chemistry, temperature monitoring, discharge rates, and combined flow/thermal modeling. Recent investigations have identified microbial communities and characterized biological activity within the springs and within permafrost sections, having direct relevance to astrobiological analogue research goals. Another main thrust of research activities based at M.A.R.S. pertains to the detection, mapping, and quantification of subsurface ice deposits. A long-term study is presently underway examining polygonal terrain, comparing surficial patterns found in the region with those identified on Mars, and using surface morphology to estimate ice wedge volumes through a combination of aerial photography interpretation and ground-based geophysical techniques. Other technological developments include the use of *in situ* microscopy for the detection of biomarkers and improved permafrost drilling techniques. This paper presents an overview of previous studies undertaken at M.A.R.S. over the past decades and will describe in detail both present and upcoming work.

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1. Introduction

Conceptually there are two approaches we can use to improve our understanding about other planets: (i) efficient analysis of existing and future data sets, and; (ii) the development of innovative techniques that could potentially be incorporated as payload or methodological constituents on future missions. With regards to Mars, one of the most effective ways to accomplish these goals is to exploit the environments on Earth that are the most similar to the conditions we expect to exist there – namely very cold and very dry – like those found in terrestrial polar deserts.

Such is the basis of the "analogue approach" to geoscientific research. Eicken (2002) notes that terrestrial analogue studies are integral to interplanetary research because they allow us to develop and test conceptual models about the properties inferred or observed on other planets and also provide constraints to theories about planetary development and composition. In essence, given the excessive costs and risks associated with full Mars missions the most viable alternative is to use to the Earth to simulate Martian environments as closely as possible.

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Table 1
Characteristics of the Canadian High Arctic making it suitable for terrestrial analogue studies.

What are desirable analogue characteristics?	Why the Canadian High Arctic?
Fidelity: Reproduce, to the maximum possible extent, the characteristics of Mars.	*Fidelity*: An isolated and harsh region of the Earth possessing a variety of Mars analogue environments.
Accessibility: Allow the analogue environment to be reached/ reproduced by CSA and potential partners (e.g. NASA).	*Accessibility*: Regular access to a limited set of facilities are provided by PCSP during the summer. Intermittent access is possible in winter.
Repeatability: Access to the analogue should be available over an extended period and under a full range of conditions.	*Repeatability*: Canadian Polar Continental Shelf Program (PCSP) has been in operation since 1958 providing high quality logistics support to Canadian scientists working in Arctic regions.
Affordability: Provide maximum leverage of available funds and partnership contributions.	*Affordability*: Relatively low-cost, because leverage of PCSP logistics costs and infrastructure is available. PCSP is interested in investing in CSA-derived technology for Arctic operations. Only area in Canada combining fidelity characteristics with available infrastructure.
Utility: If possible, provide benefits to other Government agencies, industry, isolated Communities, and scientific community by use of the analogue.	*Utility*: Provides direct benefit to PCSP work in Canadian High Arctic, and provides a terrestrial test bed to prove out systems of use to industry and to isolated communities within Canada. Will enhance use of Canadian High Arctic by scientific community.

As summarized in Table 1, the Earth's polar regions provide a unique opportunity to examine the physical and biological processes related to cold, dry environments, and to develop and test technological advances that could potentially be used for future Mars missions (Andersen et al., 1990, 1992, Vali et al., 1999). Through the Canadian Analogue Research Network (CARN) program (Hipkin et al., 2007; Osinski et al., 2007), the Canadian Space Agency (CSA) has identified two sites in the High Arctic as centers for analogue-based research: (i) the Haughton Mars Project, Haughton Impact Structure, Devon Island, and; (ii) the McGill Arctic Research Station (M.A.R.S.), Expedition Fjord, Axel Heiberg Island. The latter represents the focus of this paper.

Originally established by McGill University in 1960, M.A.R.S. (79°26′N, 90°46′W; Fig. 1) has been providing support to teams of scientists involved in research focusing on glaciology, geocryology, meteorology, biology, microbial ecology, geology and isolation psychology. While the original camp has been in operation for nearly 50 years, financial support provided through the CARN agreement with CSA has facilitated the development of a new camp approximately 10 km further west along Expedition Fjord. Additionally, M.A.R.S. field activities sometimes extend beyond the immediate local station to other parts of Axel Heiberg Island and regions of west-central Ellesmere Island such as the Fosheim Peninsula where the resources of Environment Canada's Eureka weather station are often used. Eureka is located at 79°59′N, 85°56′W, approximately 100 km east of the M.A.R.S. camp.

The primary goals of the new camp are to provide enhanced logistical support for researchers involved in CARN-funded studies, to build upon McGill's legacy of collaborations in space-related research over the past decades, and to set the foundation for future national and international collaboration in analogue research. The objective of this paper is thus to present a review of the numerous past and ongoing analogue science activities conducted at M.A.R.S. and to identify areas where future collaborations will be possible.

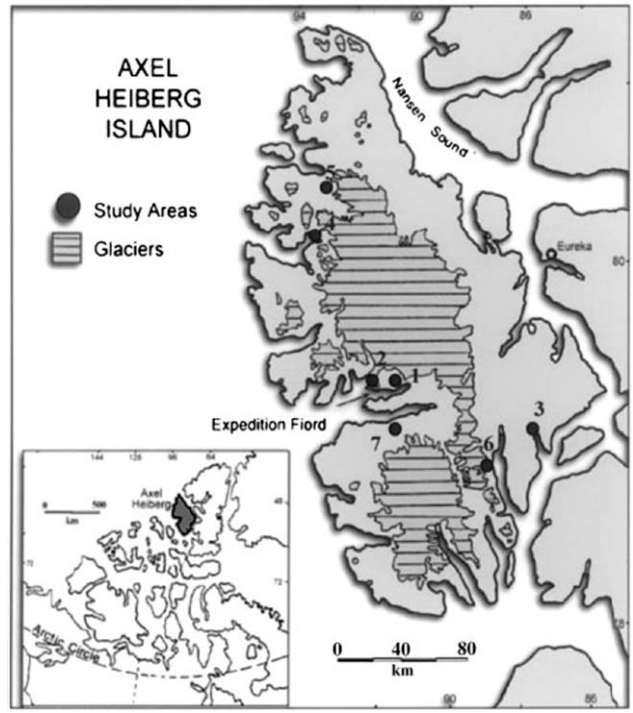

Fig. 1. Map of Axel Heiberg Island, Canadian High Arctic. Numbered study sites represent locations of perennial spring occurrences, including: (1) Gypsum Hill, (2) Colour Peak, (3) Whitsunday Bay, (4) Middle Fjord, (5) Bunde Fjord, (6) Skaere Fjord, and (7) Junction Diapir ("Lost Hammer") (See Section 3.2).

2. Physical setting

2.1. Climate

Polar desert conditions characterized by cold, dry winters and cool summers are predominant in the region. The nearest long-term meteorological records are from Eureka, which reveal a mean annual air temperature (MAAT) of −19.7 °C, mean monthly temperatures of −36.1 °C and +5.4 °C for January and July, respectively, and minimum air temperatures frequently reaching −55 °C. Periodic meteorological records are available for Expedition Fjord over the past 47 years, with a more complete record for Colour Lake available since 1992 displaying a MAAT of −15.5 °C (Andersen et al., 2008). Recent data from a broader network of automatic weather stations for the Expedition Fjord area indicate MAAT's as much 2–3 °C cooler than the Colour Lake site depending on setting.

Annual precipitation at Eureka consists of approximately 64 mm total, of which 60% falls as snow (Pollard and Bell, 1998). Though long-term precipitation values are not available for Expedition Fjord, it is assumed that the totals are somewhat greater than those measured at Eureka likely due to a rain shadow effect caused by the mountain range on the eastern Axel Heiberg Island that blocks precipitation systems from reaching Ellesmere Island (Edlund and Alt, 1989). Earlier research near Expedition Fjord suggests a mean annual accumulation of 371 mm of water equivalent on the nearby Mueller ice cap (Muller 1963).

2.2. Geology

Axel Heiberg Island is situated within the Sverdrup Basin (Hoën, 1964; Thorsteinsson and Tozier, 1970), a northeasterly striking sedimentary trough covering an area of approximately 3,13,000 km^2 (Pollard et al., 1999). Near the head of Expedition

Fjord, where the M.A.R.S. camp is located, peaks rise to a maximum of approximately 2000 m ASL. Asymmetrical ridges resulting from breached anticlines are characterized by steep scarp faces angled 70–80° and dip slopes of 25–35°.

Though piercement structures can create somewhat regular and symmetrical slope features, the area is dominated by 'serrated' profiles resulting from gypsum weathering and anhydrite outcrops along with resistant volcanic sills and dikes. The island is characterized by a series of evaporite diapirs that have been revealed by erosion over the past tens of thousands of years. The diapirs were formed by the upward intrusion of Upper Paleozoic evaporites and often appear as large domed structures cored by rock salt (Stephenson et al., 1992). The Carboniferous evaporites are comprised of an upper anhydrite layer up to 500 m thick with limestone interbeds overlying a lower layer of rock salt (Stephenson et al., 1992).

Seven perennial spring sites have been identified on Axel Heiberg Island (Pollard et al., 1999; Andersen et al., 2008). The discharge temperatures are between −4 and +12 °C and flow rates vary from <1 to 30–40 l/s. Discharge is typically mineralized with varying amounts of dissolved salts that depress their freezing temperature. The springs derive their dissolved salts from the underlying evaporates and their location most often can be linked to a nearby diapir.

2.3. Permafrost/geomorphology

Permafrost is defined as any ground material that stays below 0 °C for at least two consecutive years (van Everdingen, 2002). The depth of permafrost at the M.A.R.S. camp is estimated to be 400–500 m based on surface temperature and regional heat flow patterns. Approximately 60 km from M.A.R.S., the thickness of permafrost observed in an exploration well (oil and gas) was greater than 400 m, a value consistent with other exploration wells in the region that revealed permafrost depths 400–600 m (Taylor and Judge, 1976). The thin seasonally thawed active layer atop the permafrost typically measures 40–60 cm in thickness.

Quaternary sediments of fluvial, deltaic, marine, and glacial origins comprise the surficial deposits, with tussock microtopography dominating lower wet areas and poorly sorted circles and stripes evident at mid-elevations while bedrock dominates higher elevations. Several characteristic permafrost landforms are widespread throughout the region, including several small pingos at Middle Fjord, icing blisters and mounds in the Expedition River floodplain, and extensive polygonal terrain and ice wedge development in fluvial and colluvial deposits. Ground ice is also widespread with ice bonded permafrost occurring in unconsolidated materials, buried glacier ice occurring in most moraines, and bodies of intrasedimental ice occurring beneath fine-grained marine deposits.

3. Physical analogues

3.1. Overview

The physical conditions outlined above provide an invaluable opportunity to examine the behaviour of water in various physical states. Under the broad umbrella of the "Follow the Water" initiative (Hubbard et al., 2002) – whereby the international Mars community has identified such a task as a priority – researchers at M.A.R.S. are using Axel Heiberg Island's polar deserts as a natural laboratory in which to investigate liquid and solid water bodies found within the ground.

Introduced briefly in Section 2.2 and explained in more detail in Section 3.2, the perennial springs associated with diapirism are among the most intensively studied features on Axel Heiberg Island. By examining these saline spring systems, additional insight can be gained regarding the mechanics of icings and associated channel development. Although the low temperatures that currently characterise Mars environments and the deep cold permafrost present below its surface are the main arguments why subsurface liquid water on Mars is unlikely to exist, previous research has demonstrated that highly concentrated brines can have exceptionally low freezing points and are also remarkably susceptible to supercooling (Brass, 1980). Brines with freezing points close to the average current Martian surface temperatures (c.a. −65 °C) are found on Earth (Marion, 1997), and thus concentrated brines are excellent candidates for Martian subsurface fluids if there is a mechanism by which they could have developed on Mars.

Moreover, certain landforms that, on Earth, reflect permafrost processes have been used to suggest the presence of water ice in the Martian shallow subsurface. Specifically, patterned ground in the form of polygonal terrain can be found in mid- to high-latitudes on Mars (Mangold, 2005). Similar to those in terrestrial polar regions, Martian surficial polygons are possibly associated with ice accumulation (Seibert and Kargel, 2001). At present, few detailed comparisons of terrestrial and Martian polygonal terrain have been conducted, and there is no method of estimating the amount of ice found beneath polygonal terrain. Described in Section 3.3, a research effort has thus been focused on a variety of polygonal terrain sites near M.A.R.S. to provide a quantitative comparison of terrestrial and Martian polygonal terrains and to investigate the hypothesized relationship between surface morphology and underlying ground ice structures.

3.2. Perennial springs

At present, pure liquid water is generally considered to be unstable on Mars due to low pressure and temperature conditions (Haberle et al., 2001). However, numerous features on the Martian surface indicate the recent action of liquid water, possibly of a briny nature that would maintain an adequate vapor pressure to remain stable for short periods at the surface (McKay et al., 2005). Because the presence of a cryosphere strongly affect any hydrological activity, examining characteristics of the hydrological cycle in Earth's coldest regions may offer substantial insight with respect to Martian hydrology, past or present. Previous studies have, in fact, illustrated that liquid water could persist on Mars under modern conditions, despite mean annual temperatures that are below freezing (Andersen et al., 2002; Heldmann et al., 2005). By understanding groundwater systems in the high arctic new ideas can be developed as to how such systems could have been active on the surface of Mars in the very recent past.

In particular, perennial springs may provide relevant insight. Perennial springs are rare in regions of deep continuous permafrost because frozen ground restricts groundwater flow to either sub- or supra-permafrost regions, with the frozen material between serving as an aquitard and thus restricting exchange between the two systems (Williams and van Everdingen, 1973). High latitude perennial springs have also been reported on Svalbard and Greenland with permafrost thickness reaching 100–400 m and in some areas regional volcanics impeding permafrost formation providing a mix of environments (Lauritzen and Bottrell, 1994; Worsley and Gurney, 1996; Haldorsen and Heim, 1999). These spring environments – both cold and hydrothermal – have provided additional sites of intensive

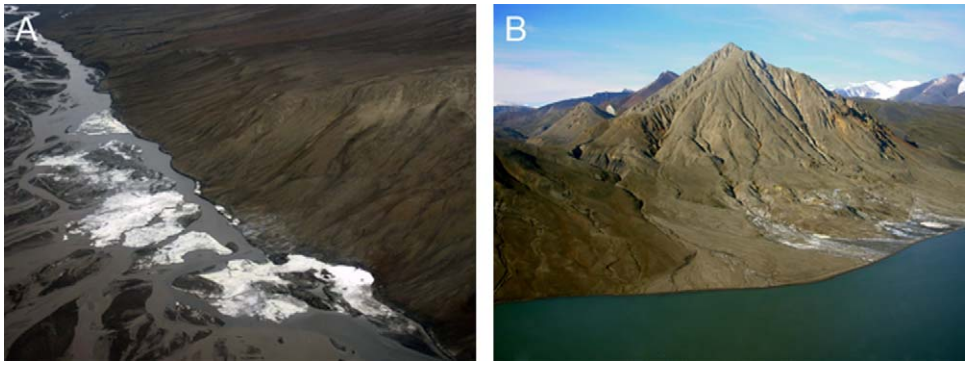

Fig. 2. Spring outlets at the base of (A) Gypsum Hill and (B) Colour Peak.

astrobiological analogue research (Amundsen et al., 2004; Steele et al., 2004).

Perhaps the most comparable spring system in a terrestrial environment is Don Juan Pond in Antarctica, where hyper-saline discharge flows throughout the year through deep continuous permafrost (Marion, 1997). Similarly, springs on Axel Heiberg Island demonstrate year-round liquid discharge of virtually constant flow rates and discharge temperatures, despite the fact that winter air temperatures frequently reach minima in the vicinity of −50 °C (Pollard et al., 1999; Andersen et al., 2002).

The first description of perennial spring activity near the M.A.R.S. camp was provided by Beschel (1963), who noted the presence of a group of seeps roughly 2 km away from the main camp at the base of the south facing slope formed by the Expedition Diapir (Gypsum Hill; 79°24′30″N, 90°43′05″W; Fig. 2A). Though mentioned in various papers over the following three decades (Allan et al., 1987; Schiff et al., 1991), this hydrological system remained largely unstudied until Pollard (1991) returned to the site to investigate the springs further. Described by Pollard et al. (1999), over 40 individual springs and seeps are present at this site, discharging over a band nearly 300 m long and 30 m wide at the bottom of Gypsum Hill where a sharp break in slope is coincident with an overlap of bouldery colluvial material and sandy outwash deposits. Total discharge for all of the outlets is approximately 15–20 l/s, the average outlet temperature (weighted by volume) is 2 °C, and the chemistry of the solution is dominated by dissolved Na and Cl with lesser amounts of K, Ca, Mg, and SO_4 (Pollard et al., 1999).

The outlets at Gypsum Hill are randomly spaced over an area of 2000–2500 m^2 adjacent to the Expedition River, a large glacier floodplain fed by meltwater from the White and Thompson Glaciers. Springs outlets range from well-defined vents and pools to patches of saturated sediment. The outflow forms small gypsum and calcium carbonate coated channels and flow paths that cascade roughly 15 m to the Expedition River. Any features occurring on the Expedition flood plain are removed annually by glacial meltwater. Flow rates and temperatures are constant year round. In winter a thin veneer of snow covers some of the small higher outflows while icings and icing mounds form further downslope. A large icing (0.3–2.1 m thick) forms every year extending from the spring outlets at the base of Gypsum Hill to 700 m down stream and up to 300 m out onto the floodplain. Snowmelt and water from the active layer contributes very little to flow. Salt and gypsum precipitates cover much of the active spring area (Omelon et al., 2006a).

Over the following years, it was discovered that the Gypsum Hill springs were not an entirely unique occurrence on Axel Heiberg Island. A second set of springs was reported by Pollard and Bell (1998) and Pollard et al. (1999), approximately 11 km from Gypsum Hill towards the mouth of Expedition Fjord at the base of Colour Peak (79°22′48″N, 91°16′24″W; Fig. 2B), a heavily eroded and incised 560 m high pyramid-shaped mountain formed by Colour Diapir.

Similar to the Gypsum Hill location, the spring system at Colour Peak is characterized by multiple outlets; in the case of the latter, approximately 30 springs discharge over an area of 2300 m^2 and range in elevation from 10 to 90 m asl. These springs can be grouped into three distinct zones, with both the east and west zones consisting of four to seven outflows within a V-shaped channel 5–8 m deep that collects the discharge and directs it into Expedition Fjord. Outflows are marked by small vents and seepage zones. The middle zone consists of more than 15 outflows marked by distinct vents and pools. Hard carbonate trough structures and travertine-like terracetes that cascade from the spring outflows downslope to the edge of the fjord make the middle zone very distinctive. These carbonate structures tend to separate the flow from each spring. A salt efflorescence coats much of the Colour Peak spring area. Like Gypsum Hill the snow cover is very thin and contributes very little to the flow. At both sites flow meters and temperature sensors show very little change during the nival freshet. However in both cases the melt of icings deposits that accumulated over the winter may add to the downstream flow. Both spring groups are marked by a discharge of foul smelling gases.

At Colour Peak, the erosional potential of the springs have formed a series of deep gullies near the base of the slope, where a combination of weathered bedrock and precipitate-covered silty mud comprise the surficial material. Total discharge is estimated at 20–25 l/s and average discharge temperature is approximately 6 °C (Pollard et al., 1999). While the chemistry of the water is similar in dissolved ionic content to Gypsum Hill, only the springs at Colour Peak result in the periodic formation of the quasi-stable mineral ikaite (Omelon et al., 2001).

Although the physical and chemical characteristics of the outflows at these two sites had been well characterized, the source of the water was as of yet unknown. While Pollard et al. (1999) had originally presented five possibilities, Andersen et al. (2002) developed a combined flow and thermal model to demonstrate that the brine discharge may originate in a glacially dammed lake several kilometers away. The mechanism that the authors propose involves drainage of lake water through a sub-lake talik into a sub-permafrost evaporite layer, and the return to the surface through the evaporite piercement structures with which the springs are associated. As noted by Andersen et al. (2002), such a model could then used to propose a method by which possible Martian brines could have been brought to the surface to initiate erosive action.

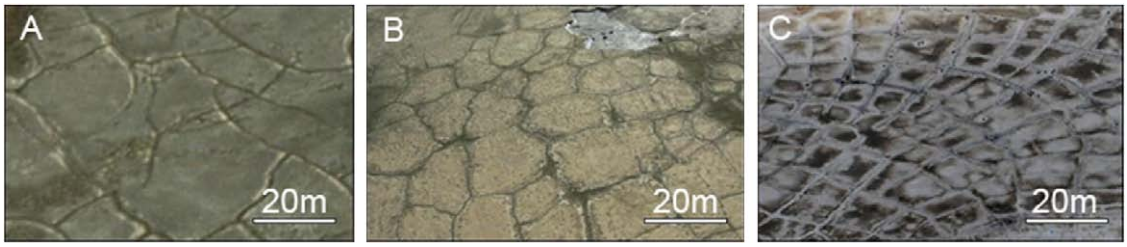

Fig. 3. Variety of polygonal terrain morphologies in the Canadian High Arctic.

A number of ideas have been put forth to explain the presence of gully systems reported on Mars. These range from simple erosion due to the melt of snowpack or ice, to the flow of liquid CO_2, to simple mass wasting (Costard et al., 2002; Malin et al., 2006; McEwen et al., 2007; Pelletier et al., 2008). The notion of spring activity being responsible for gully erosion on Mars was further examined by Heldmann et al. (2005). Here, the authors presented a numerical model illustrating that current Martian environmental conditions may be sufficient to allow short-term residence of pure liquid water to carve the Martian gullies described by Heldmann and Mellon (2004). The length scales predicted by the model of Heldmann et al. (2005) match well with the outflow distances observed at Gypsum Hill, and further demonstrate that pure liquid water (as opposed to brines or ice-laden flows) may be a possible erosive agent responsible for Martian gullies.

Though research to date has focused primarily on the Colour Peak and Gypsum Hill springs, recent reconnaissance missions have revealed the presence of five additional spring systems in the vicinity (Fig. 1; see Andersen et al. (2008) for an extensive review). While research on the hydrology and geomorphology of these features is ongoing, subsequent investigations have illustrated a great diversity in microbial ecology within the springs (e.g. Perreault et al., 2007). If life on Mars was to have ever arisen, it would most likely have been associated with groundwater systems (McKay et al., 2005) and thus it is possible that these springs may also have significant astrobiological implications. These notions are elaborated upon in detail in Section 4.2.

3.3. Polygonal terrain

Polygonal terrain refers to a network of interconnected trough-like depressions in the ground that form enclosed geometrical shapes with spatial dimensions on the order of meters to tens of meters across (Fig. 3). In terrestrial polar regions, these features are often diagnostic of underlying ground ice deposits in the form of ice wedges (French, 1996).

As such, the discovery of comparable surface features on Mars has led to their being interpreted as analogous landforms that may also potentially be indicative of subsurface ice (Seibert and Kargel, 2001). The most common (and most obvious) comparison to be made between terrestrial and Martian polygonal terrain is the observed similarities of the geometric patterns created by these landforms (Isaev and Abramenko, 2003; Mangold et al., 2004). However, few examples in the literature pertain to quantitative comparisons of the two (Rossbacher, 1986); rather, the noted similarities are often based on qualitative interpretation and description (e.g. Mangold, 2005).

The primary difficulty in comparing geometrical characteristics of terrestrial and Martian polygonal terrain is that no common quantitative classification scheme currently exists. Through the CSA–CARN program, a long-term study has commenced at M.A.R.S. to examine certain statistical properties of polygonal geometry and to interpret intra- and inter-site variations based on geomorphological principles.

The underlying statistical foundation of this work is based upon spatial point pattern analysis (SPPA), a technique described in detail by Diggle (2003) and summarized briefly below. SPPA examines the spatial distribution of identifiable 'node points' as being random, regular, or aggregated, and provides information about the distances between the observed nearest neighbour nodes. Under the test's classical null hypothesis, an envelope curve representing a simulated "random" distribution is defined and the observed nearest neighbour distances are superimposed upon the plot. If the observed nearest neighbours lie within the envelope curve, the null hypothesis is accepted and the observed points are considered to be randomly distributed; if they do not lie within the envelope, they can be interpreted as being either regular or aggregated.

Dutilleul et al. (2008) describe the statistical procedure of applying SPPA to terrestrial and Martian polygonal patterns, whereby polygon trough intersections are used as the 'node points'. A selection of Mars Orbiter Camera (MOC) images and aerial photos of test sites near the M.A.R.S. camp – all of which reflect qualitative variations in polygonal geometry – were used to illustrate that SPPA is an effective means of quantifying the apparent inter-site morphological variation. Ultimately, the authors propose that this technique could potentially be used as a building block to develop a common classification scheme for more objectively and rigorously comparing polygonal terrain on Earth and Mars.

Haltigin et al. (2007) provided a preliminary interpretation of SPPA-derived variations in polygonal geometry based on the sites' rheological characteristics. Specifically, the authors show that the test sites on Axel Heiberg Island display variations not only in polygonal geometry but also in surface sediment size distributions, demonstrating a correlation between substrate heterogeneity and observed polygon network randomness. Given that thermal contraction cracks leading to the development of polygonal terrain initiate at weakness points throughout the site (Lachenbruch, 1962), it is plausible that a more heterogeneous substrate would have more randomly distributed weaknesses, which in turn would be reflected in a more random spatial distribution of crack intersections. These notions will be expanded upon in an upcoming manuscript.

Keeping these principles in mind, SPPA has been used to propose explanations for geometrical variations observed in polygonal geometry on Mars. For example, Haltigin et al. (2008a) showed that the morphologies of polygonal terrain sites near the Mars Phoenix landing area are correlated with surficial geomorphic units. Because the various units differ in albedo, which in turn varies partly due to differences in surface material, it is possible that a demonstrable relationship between substrate type and polygonal morphology is applicable to this region of Mars.

Terrestrial polygonal geometry is also dependent upon surrounding climate conditions (Mackay, 1993) and surface age (Sletten et al., 2003). Similarly, it is possible that these factors also affect Martian polygon network geometries. Haltigin et al. (2008b) used sites located throughout western Utopia Planitia to illustrate that variations in polygon geometry throughout the region are partially dependent on the sites' elevation and location, which in turn may reflect differences in both climate and surface age. Pertaining to the statistical basis of this relationship, the degree to which observed spatial point patterns of terrestrial polygon networks are affected by surface age is currently under investigation at the test sites near M.A.R.S. and others throughout the Canadian High Arctic.

The long-term objective of the CARN study on terrestrial polygonal terrain sites is to investigate the relationship between surface morphology and underlying ice wedge dimensions, with the ultimate goal of developing an algorithm whereby high-resolution satellite and/or aerial photos can be used to estimate ice wedge volumes. While the surface interpretation is founded upon spatial point patterns and other image analysis techniques, a project is currently underway to use complementary ground-based geophysical tools to profile subsurface characteristics at the sites (see Section 5.2). Upon completion, it is hoped that the relationships between surface morphology and ground ice volumes can be extrapolated and translated to Mars using HiRISE imagery to provide a first order estimate of potential ground ice volumes found beneath Martian polygonal terrain.

4. Biological analogues

4.1. Overview

Various studies have described the microbial communities inhabiting Earth's diverse environments characterized by low or sub-zero temperatures (de la Torre et al., 2003; Stackebrandt et al., 2004; Gilichinsky et al., 2005; Barrett et al., 2006; Mikucki and Priscu, 2007; Perreault et al., 2007; Steven et al., 2007a), with many of these communities being metabolically active under these sub-zero *in situ* conditions (Carpenter et al., 2000; Rivkina et al., 2000; Christner, 2002; Junge et al., 2004; Steven et al., 2007b; Wagner et al., 2007; Perreault et al., 2008). This holds important astrobiological implications as the settings of these habitats overlap to some degree with the terrestrial environment of Mars (Baker, 2001).

Multiple projects are underway investigating the abundance, distribution and phylogenetic/physiological composition of the microbial communities inhabiting several unique cryo-habitats of the Canadian High Arctic. In another study, endolithic communities of blue-green algae that live immediately below the surface of coarse-grained sandstones are being examined. This research also examines the role of these cryptoendoliths in rock weathering. Sections 4.2 and 4.3 present the results of these works, describe the utility of these cryo-environments as extraterrestrial analogues, and outline future research projects.

4.2. Microbial ecology

4.2.1. Gypsum Hill and Colour Peak

As outlined in Section 3.2, the perennial springs of Gypsum Hill and Colour Peak may provide useful analogues for Martian hydrological systems (Andersen et al., 2002) and present a novel environment for the investigation of chemosynthetic communities (Perreault et al., 2007, 2008).

Culture- and DNA-based investigations revealed that the microbial communities inhabiting sediments of springs at both locations (Fig. 4A) are dominated by signatures being related to potential sulfur-metabolizing bacteria. Thus, it was postulated that sulfur-based metabolism may be the major source of energy production in the sediments of these springs (Perreault et al., 2007, 2008). Patchy distributions of grayish-white biomass had also been observed in the runoff channels originating from the springs at GH during previous summer month expeditions. (Fig. 4B). However, has never been observed in the springs at Colour Peak. During a late winter expedition of 2007, the removal of the snow covering the perennial runoff channels of the Gypsum Hill springs revealed an increase in the amount of biomass as compared to the summer months.

Analyses undertaken on the biomass indicated that the microbial population is dominated by a genus of the *Gammaproteobacteria*, namely *Thiomicrospira* (Niederberger et al., 2008) and the biomass was proven to undertake both sulfide and thiosulfate oxidation and CO_2 uptake at *in situ* conditions, which is consistent with the presence and activity of chemolithoautotrophic sulfur-oxidizing bacteria such as *Thiomicrospira*. It is presumed that the S-oxidizing microbial community flourishes under the winter conditions due to the trapping of their presumed energy source, H_2S, beneath the snow covering the runoff channels. S-based chemolithotrophy has been shown to sustain microbial communities in ecosystems devoid of light such as hydrothermal vents (Jannasch and Mottl, 1985) and this non-photosynthesis-based primary production would also permit to sustain the spring microbial communities during the months of total darkness that occur seasonally at these high latitudes. As such, the fact that these microbial structures can form and flourish under sub-zero temperatures via chemolithotrophic, phototrophic-independent means is of interest in astrobiology, particularly for the research of subsurface waters that are hypothesized to exist on Mars.

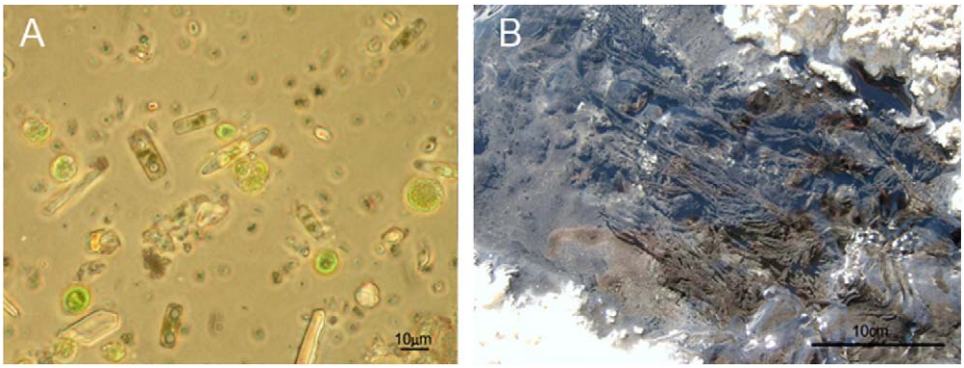

Fig. 4. (A) Phase contrast image of the microbial community (bacteria, protozoa, diatoms, algal cells) found in cold saline spring channel sediments (scale bar approximately 10 μ); (B) Sulfur-oxidizing microbial filaments in a Gypsum Hill cold saline spring runoff channel (scale bar approximately 10 cm).

4.2.2. Lost Hammer

Geochemical analysis of the Lost Hammer (LH) spring indicates that it is an extreme environment with an ambient temperature of −4.8 °C and a salt concentration of 22–23% and is emitting methane, CO_2 and H_2S. Microscopic-based analyses indicate the presence of a viable ($\sim 4.4 \times 10^5$ cells/g wet Lost Hammer sediment) microbial community that consists of both bacteria and archaea. The majority of the archaeal signatures were related to the *an*aerobic *me*thane group 1a (ANME-1a) of archaea with the closest relatives being signatures detected in marine methane-seep sediments (Hinrichs et al., 1999). Bacterial phylotypes were in general strongly related to microbes previously detected in environments of Antarctica and the Arctic (Bowman and Nichols, 2005; Nedashkovskaya et al., 2005; Brinkmeyer et al., 2003; Van Trappen et al., 2004a,b).

Isolates cultured from this spring have proven to be active under extreme salinities and temperatures (>20% NaCl and −10 °C) in the laboratory setting and low heterotrophic activities were also detected in Lost Hammer LH sediment at temperatures as low as −10 °C (Steven et al., 2007b). Therefore, Lost Hammer is remarkably interesting from an ecological perspective as the possibility that life can exist in such an extreme, highly saline, sub-zero environment has implications not only on the limits defining life on Earth but also life on other planets, in particular Mars.

As part of the above-mentioned studies, a large number of novel bacteria (>200) have been isolated from these sites. As part of an upcoming project, a number of these isolates will be screened for species tolerant to extreme conditions and the survivors will be incubated in a Mars environmental simulator (SHOT Inc., Facility, Indiana, USA) to test the survivability and potential activity of these strains under Martian conditions. It is hoped that the knowledge from these studies will help us understand the potential risk of forward contamination of Mars via future exploration missions.

The significant finding that microbial respiration in Eureka permafrost and Lost Hammer sediment can occur in the laboratory setting at temperatures as low as −15 °C and −10 °C (Steven et al., 2007b) indicates that similar microbial activity could potentially occur *in situ* at ambient high Arctic conditions.

In 2007, the use of a highly sensitive and portable CO_2 flux detection system (LI-8100, LI-COR Biosciences) indicated very low but detectable CO_2 fluxes at sub-zero conditions from permafrost soils near M.A.R.S. and the cold saline spring sediments on Axel Heiberg Island during the late winter months. In the coming years, we hope to determine if the CO_2 detected is from a biotic (microbial respiration) source or rather natural abiotic CO_2 flux. This will be investigated through collaboration with the Onstott team from Princeton University involving the use of a portable cavity ring down spectroscopy (CRDS) system (Onstott et al., 2006). The CRDS being developed by Onstott's team is a portable, highly sensitive gas detection system that could determine the isotopic variation and potential origin of the CO_2 released from sites at Axel Heiberg Island. Collectively, the fundamental knowledge from these investigations is crucial to understanding the limits defining life on Earth and possible microbial life in similar Martian habitats and could lead to the development of *in situ* robotic methodologies to detect such life on Mars or Europa.

4.3. Cryptoendolithic habitats

Endolithic microorganisms (including bacteria, eukarya, and archea) occupy a unique niche in terrestrial habitats by living beneath rock surfaces as interstitial habitants of cracks, fissures, and pore spaces between mineral grains (Goublic et al., 1981) (Fig. 5). These microbial communities are of wide interest to studies in astrobiology and geomicrobiology as they are found at the organic–inorganic interface and have profound implications for our understanding of dissolution and precipitation reactions that lead to both the destruction of lithic substrates as well as the creation of authigenic minerals to produce biosignatures. Recent discoveries indicating that widespread liquid water likely existed on Mars in the past has heightened scientific interest in determining the potential for life to have once existed on that planet. Endolithic microbial communities in cold desert environments define the limits of life in polar regions, and have repeatedly been suggested to be one of the most likely places for viable life on Mars as it entered its final cooling period (Friedmann and Koriem, 1989; Friedmann, 1982; Mackay, 1999). Current interest in astrobiology has led to an improved capacity to identify biosignature preservation in a variety of natural environments, and endolithic communities in polar deserts are a favourable research focus for Mars analogue studies.

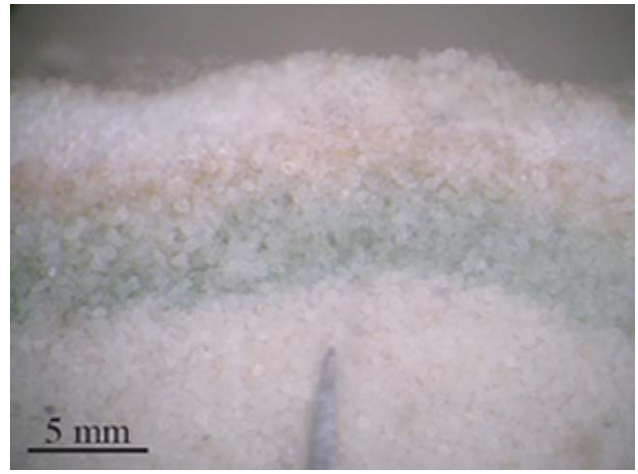

Fig. 5. Dark band of cryptoendolithic microorganisms millimeters beneath the rock surface.

Endolithic microorganisms are found in local or regional areas where climatic extremes limit epilithic colonization of rock faces, establishing communities only millimeters beneath exposed surfaces where they are protected from environmental stresses such as extremes in temperature, aridity, radiation and winds, but able to obtain necessary nutrient, moisture and light requirements for survival. Friedmann and Ocampo (1976) reported the first occurrence of polar microbial communities in cryptoendolithic habitats in the McMurdo Dry Valleys of Antarctica (77°36′S, 161°05′E). The discovery of this ecosystem was recognized as important to the study of extreme life forms as they were the only observable near-surface terrestrial microbial activity in one of the most inhospitable environments on Earth (MAAT <−30 °C, precipitation rates <10 mm yr^{-1}), making it an ideal focus for Mars analogue studies (Friedmann and Ocampo-Friedmann, 1984; Friedmann, 1986; Friedmann, et al., 1986; Friedmann and Koriem, 1989; Wynn-Williams and Edwards, 2000a, b; Wierzchos and Ascaso, 2002; Wierzchos et al., 2003, 2005).

The more recent discovery of cryptoendolithic microorganisms around Eureka, Ellesmere Island, Nunavut in the Canadian High Arctic in 2001 led to the development of a bi-polar comparison between these microbial communities and their environments (for a recent review of these comparisons see Omelon, 2008). Documentation of microenvironmental conditions within endolithic habitats (McKay and Friedmann, 1985; Friedmann et al., 1987; Omelon et al., 2006b) revealed how a combination of warmer temperatures and elevated moisture

levels in the subsurface creates conditions amenable to microbial colonization, with warmer temperatures and abundant moisture in the form of liquid water during summer months in the Arctic leading to longer time periods for metabolic activity (Omelon et al., 2006b).

Investigations of biosignature formation observed as microfossils produced conflicting results between these two locations: in contrast to the abundance of microbial fossils in cryptoendolithic habitats in the Dry Valleys of Antarctica (Wierzchos et al., 2003, 2005), no evidence for biomobilization and biotransformation of inorganic deposits to produce diagenetic biomarkers nor mineral encrustation and infilling of these microorganisms to produce biosignatures and microfossils are observed in this area of the Canadian High Arctic (Omelon et al., 2006c). This is believed to be a function of warmer temperatures and a greater abundance of liquid water that promote faster rates of biomass and element cycling and, more importantly, promote destruction of the cryptoendolithic habitat and subsequent exfoliation of the overlying rock (Friedmann and Weed, 1987; Kappen, 1993; Sun and Friedmann, 1999) thereby exposing the microbial community to more harsh aerial conditions and their removal by winds (Friedmann et al., 1987; Friedmann and Weed, 1987; Omelon et al., 2006b).

Variations in microbial community structure corresponding to chemical differences within cryptoendolithic sandstone habitats were found both in the Canadian High Arctic and the Antarctic Dry Valleys (Johnston and Vestal, 1989; Omelon et al., 2007) and is likely a reflection of the activity of microorganisms (Blackhurst et al., 2004). The generation of high pH conditions by cyanobacteria possessing a carbon concentrating mechanism (Omelon et al., 2008) leads not only to the creation of unique exfoliation patterns on rock surfaces (indirect biosignatures) due to accelerated silicate weathering, but elevated concentrations of calcium and magnesium ions observed in the presence of these microorganisms highlights the potential for precipitation of calcium and magnesium carbonates on cell surfaces (direct biosignatures) that may be preserved within the rock record (Omelon et al., 2007).

Investigations into answering the question as to what controls microbial diversity in these habitats suggests that rock type alone cannot explain observed variations in community composition both in the high Arctic and other locations, suggesting that site-specific characteristics such as microclimate, geochemistry, or surface crust formation may have a stronger influence in controlling species diversity than large-scale (regional) conditions (Omelon et al., 2007; Walker and Pace, 2007; Omelon, 2008).

Ongoing controversy over our ability to positively identify microbial fossils highlights the problematic nature of this scientific discipline (Brasier et al., 2007). Most notable in this regard is a poor understanding of the *processes* involved in microbial biomineralization and fossilization, and their subsequent preservation in the rock record. Ongoing work focuses on cyanobacteria in these high Arctic endolithic habitats and identifying both evidence and mechanisms for authigenic mineral formation (carbon concentrating mechanisms, carbonic anhydrase activity, cell surface reactivity, and mineral precipitation/dissolution reactions at the microbe/mineral interface), and will provide new information about how, why, and to what degree these microorganisms are involved in biosignature formation and preservation both near Eureka and at other terrestrial analogue study locations.

5. Technological development

5.1. Overview

Traditionally the research thrust at M.A.R.S. has been dominated by scientific investigation with a focus on understanding particular characteristics or functions of physical and/or biological analogue systems. Recently, however, a series of projects have commenced focusing on technological development for exploration purposes, comprised of three stages. First, "off-the-shelf" commercially available technology is being tested to conduct proof-of-concept studies, designed to illustrate the utility of these instruments for a particular scientific investigation. After preliminary usage, the second step is to propose and effect initial modifications to the commercial instruments so as to refine their use in the harsh cold and dry environments present in the high arctic. Finally, new instrumentation is being designed and constructed with the explicit goals of miniaturization and reduction of power consumption such that these prototypes could potentially be incorporated on future Mars missions.

Presently, researchers at M.A.R.S. are conducting technological studies in each of these three phases. As outlined in Section 5.2, off-the-shelf geophysical instruments are being used to illustrate that the detection of subsurface ice bodies is greatly enhanced by combining multiple complementary datasets. Described in Section 5.3, a commercially available permafrost drill has been modified to improve sample retrieval and to examine potential contamination issues for future Mars drills. Finally, Section 5.4 provides an explanation of efforts to utilize *in situ* microscopy for the detection of biological materials using miniaturized prototype instruments.

5.2. Geophysics

Non-invasive geophysical techniques are widely used for ground ice and permafrost-related field investigations. Namely, ground penetrating radar (GPR) is used to determine subsurface structure, providing a stratigraphic reconstruction based on differences in the ground materials' dielectric properties (e.g. Hinkel et al., 2001; Arcone et al., 2002; Fortier and Allard, 2004). Another method – capacitive coupled resistivity (CCR) – is used to identify subsurface materials based on the calculation of electrical resistance of the materials through which the induced signal passes (e.g. Calvert et al., 2001; Calvert, 2002). Conceptually similar to resistivity, electromagnetic profiling (EM) identifies near-subsurface materials based on the substrate's electrical conductivity, the mathematical inverse of resistivity.

Given the respective strengths of each method, it is clear that a great advantage would be gained by combining these types of data. Although each method is reasonably well-established individually in permafrost environments, little effort has been placed on combining them to produce a more detailed subsurface interpretation.

The potential of multi-tool geophysical surveys for assessing ground ice volume and structure on Mars must be examined. Because GPR, CCR, and EM are all candidate payloads for future Mars rovers, it is imperative that we use terrestrial analogue environments to develop the techniques required to interpret geophysical data on known ice deposits. In essence, it is important to investigate the utility of these tools individually and in a complementary fashion under diverse climatic and geomorphic conditions to examine the extent to which these tools are able to detect and quantify ground ice deposits.

Previously, EM and GPR were thought to have the greatest potential for Mars exploration applications; based on our experience with CCR, though, we believe that it may be advantageous to combine CCR with GPR and EM. This is the theoretical assumption that formed the basis of the *Resistivity Instrumentation for Ground Ice Detection* (RIGID) concept study. As a test of this assumption the utility of GPR and CCR individually and in complementary fashion under diverse climatic and

geomorphic conditions to detect and assess ground ice was investigated.

Preliminary fieldwork on combining GPR and CCR for the purposes of ground ice detection was conducted in the Canadian Western Arctic. Williams et al. (2005) introduced the utility of a Mars-prototype GPR at detecting the base of the seasonally thawed ("active") layer and detection of massive ice bodies, illustrating that the GPR signal is affected by variations in the physical properties of frozen ground and massive ice and noting that the interpretation of a returned GPR signal is aided greatly by the inclusion of CCR data.

DePascale et al. (2008) subsequently presented the first comprehensive case study of combining CCR and GPR in permafrost environments, demonstrating clearly that both dielectric (GPR) and resistivity (CCR) transitions in the returned data can be used to interpret stratigraphic contacts between massive ice and other frozen materials. Moreover, the authors show that these instruments are extremely useful for detecting and imaging various types of ground ice, including tabular massive ice and wedge ice.

We have recently continued examining the applicability of complementary geophysical tools at various sites near M.A.R.S. For instance, preliminary GPR surveys were conducted near the Colour Peak springs. These data reveal that the increased salinity of the subsurface associated with spring activity greatly increases the ground's dielectric constant and thereby strongly attenuates the GPR signal (Haltigin and Andersen, unpublished field data).

Additionally, through CARN initiatives an increased effort in mapping ice wedge volumes beneath polygonal terrain is also underway. In July 2007, numerous transects over wedges at a variety of sites were surveyed using GPR (operating at 200 and 400 MHz), CCR, and EM-31 instruments. Preliminary results show a strong relation amongst the three datasets, with decreases in near-subsurface conductivity (EM) being spatially coincident with increases in subsurface resistivity (CCR) and decreases in dielectric permittivity (GPR) as would be expected over massive ground ice deposits (Milsom, 2003). These findings will be discussed in several upcoming manuscripts.

5.3. Drilling

As outlined in Section 4, microbial life has proven extremely resilient and has shown that it can both survive and thrive in cold environments previously thought uninhabitable on Earth. It follows, therefore, that extremely cold and dry terrestrial sites with ice-rich permafrost and thick unconsolidated surficial geology are important potential analogue targets where questions about ice origin, age, stability and microbial ecology can be addressed.

A variety of studies have punctuated this point, showing that viable microbial communities are present within the permafrost itself (Steven et al., 2007a). Although there remain many questions concerning the diversity and level of activity of bacteria in cold permafrost and their relationship with ground ice, our preliminary findings (Steven et al., 2007a) show not only that living bacteria survive in permafrost $<-10\,^{\circ}C$ but that they preferentially occur along pore ice-sediment grain boundaries. Therefore, it is important to refine the methods – especially with regards to contamination-free "clean drilling" (e.g. Juck et al., 2005) – whereby subsurface samples can be collected for subsequent analysis. The overarching goals of our drilling activities are the acquisition and biophysical analysis of near-surface ground ice and ice-rich permafrost samples. Specific analytical objectives include: (i) determination of ice content, origin, and age; (ii) determination of living and preserved microorganisms, and; (iii) the determination and characterization of bacterial microhabitats.

This work is an outgrowth of our participation in the NASA Ames/JSC/Baker Hughes ASTID funded Mars Autonomous Deep Drill (MADD) project. Although permafrost drilling is an important component of our project, the development of Mars drilling technology is not an explicit objective. However, we are hopeful that lessons learned concerning sample recovery, handling, and contamination will be useful in future Mars drill designs.

Sample acquisition involves permafrost coring based on a modified CCREL/SIPRE corer. This system typically consists of a power head (3–5 hp) with a low rpm automatic transmission geared to maximize torque, light weight drill rods (~1 m long) that are manually connected with depth and a specially designed CRREL 3″ core barrel (18″ long) with carbide cutters. Unlike the automated MADD, the modified CCREL system is usually operated by two persons who physically hold the power head and control the throttle.

Cores have been obtained at three locations to depths ranging between 1.7 and 3.0 m, with massive ice and ice-rich sediments having been encountered at depths of 90–110 cm. Each sample was logged and sub-sampled for ice content. Visual estimates of volumetric ice contents between 30% and 79% were confirmed when samples were field weighed and dried. A Bodelin Technologies ProScopeHR digital microscope with $\times 100$ and $\times 400$ magnification was used in the field to assess core structure.

Pore spaces were found to be ice filled and individual soil grains had ice coatings. Unlike non-permafrost soils where pore spaces provide open cavities for bacterial activity and moisture migration, permafrost soils are largely impermeable due to the presence of ice. By painting the core with fluorescent microspheres it was possible to determine that the most likely microhabitats occur along intercrystalline boundaries and in association with brine films (Juck et al., 2005). Subsequent laboratory analyses (ongoing) have yielded new and interesting data on the nature and diversity of Bacteria and Archaea.

Future activities include the design and construction of a system based on a prototype construct by the Geological Survey of Canada. In this case the power head is mounted in a track on a 2.5 m mast and is raised and lowered using a cable/chain and geared to be able to add down-hole pressure to the cutters. The mast can be mounted on either an ATV or a cargo sled. Ideally we will be able to core to depths of 3–5 m. During drilling we will experiment with a range of tracers (e.g. fluorescent microspheres) to assess aseptic drilling requirements evaluate potential biochemical contamination sources.

5.4. Microscopy

The availability of a small, robust fluorescence microscope capable of distinguishing individual microorganisms would be of great value to many microbiologists working in extreme environments. Currently, researchers traveling to the Arctic and Antarctic cannot identify or count microbes in their samples before returning them to their laboratories, which creates significant expense in both shipping of samples and repeat trips to remote locations, as well as loss of time due to the seasonal unavailability of the sites.

Such an instrument would also be useful in space applications (Kawasaki, 1999; Cady et al., 2003). Used alone as part of a Mars lander, a miniaturized fluorescence microscope would be able to unambiguously identify certain minerals and provide strong evidence of the existence (or absence) of fluorescent bacterial pigments, lipids, polysaccharides, and other cellular components. Used as part of a complete wet chemistry suite, high-resolution

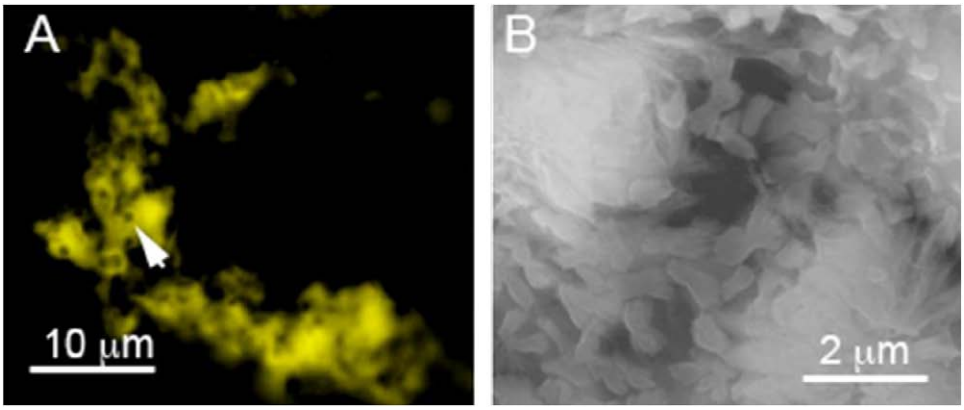

Fig. 6. Imaging of mineralized biofilms from the Canadian High Arctic using a Gram-negative specific QD probe. (A) Confocal laser scanning image showing outlines suggestive of bacteria; (B) Environmental scanning electron micrograph at higher resolution, showing organisms adhering to the sulfur minerals.

imaging may provide one of the best methods for identification of extinct or extant microbial communities on Mars.

The goal of our work is to create an instrument for *in situ* fluorescence microscopy that is suitable for harsh environments and that will eventually enable completely robotic staining and imaging of microorganisms in sediments, in pools, and on rocks. The instrument will consist not only of a microscope, but also of a wet chemistry suite allowing delivery of sample, staining with selected fluorescent probes, and automated focusing and imaging.

The Mars Exploration Program Analysis Group identified microscopic imaging as an essential element for the contact suite of a Rover mission. The next landed mission to Mars, Mars Science Laboratory (MSL), contains a geological hand lens (MAHLI) with a resolution of 12.5 μm/pixel and true-color capability. Future missions will thus begin with optical data on Martian regolith on the tens of microns scale. Improving on this resolution has until now been unattainable, as high-resolution microscopes are heavy and their optical components are fragile.

Spectroscopic instruments in the infrared and X-ray have also been selected for Martian missions both past and future. Such spectra allow for identification of minerals and have been of enormous importance in determining the extent of liquid water in Mars' past. However, visible fluorescence data have yet to be collected. All minerals show fluorescence or luminescence, and visible fluorescent examination can provide clues to the deposition of salts and the location of mineral defects, as well as confirmation of mineral identification. The ultraviolet LED arrays on MAHLI will also provide early data on the value of this approach, although they may not be powerful enough to excite most mineral fluorescence.

A high-energy fluorescence/transmission microscope system capable of single-micron resolution would thus be able to expand upon current goals of observing Martian regolith grain shapes and textures, determination of grain sizes, and identification of sedimentary deposits. Combined with state-of-the art methods of fluorescent labeling, it could also serve as a powerful instrument for life detection, as fluorescence microscopy is the gold standard for identification and enumeration of living bacteria on Earth.

The work consists of three elements: (i) identification and development of fluorescent probes compatible with *in situ* instruments; (ii) refinement of a microscope design, and; (iii) packaging of the entire suite within a wet chemistry instrument. All of these elements are tested in the Canadian High Arctic as well as in the laboratory using samples collected from a wide variety of extreme environments, including the McMurdo Dry Valleys and the Atacama Desert.

Development of fluorescent probes in our laboratory has primarily focused on semiconductor nanocrystals or quantum dots (QDs) (Kloepfer et al., 2003, 2005). These particles have many desirable properties for *in situ* instrumentation, such as broad absorption spectra allowing them to be excited by nearly any visible or near UV light source; narrow emission spectra, allowing for multicolour labelling; and resistance to radiation, aging, and freeze-thaw (Bruchez et al., 1998; Nadeau et al., 2008). We have developed QD probes that are specific for Gram positive and Gram negative bacteria, and which can be used directly on unprocessed, living environmental samples for both fluorescence and electron microscopy (Fig. 6).

Two types of microscopes are currently being developed and tested. The first is an ultraminiaturized instrument ($20 \times 20 \times 10$ mm^3) based upon a miniature optical table (MOT) platform fabricated using deep X-ray lithography (Rogers et al., 2004). It is capable of reflection, transmission, and epifluorescence imaging, although its current resolution is still too low for any but the largest bacterial cells in pure culture. Its fluorescence detection capability is also weak and restricted to the red (>630 nm), allowing us to image quantum dots but no other tested dyes or autofluorescence.

The second microscope is a meso-scale, robust instrument containing all of the elements of a desktop fluorescence microscope. Size is reduced by limiting the optics (one objective lens; one bandpass excitation filter and one longpass emission filter) and by replacing the high-power mercury excitation lamp with light-emitting diodes. While too large to be packaged into a flight instrument, this microscope has permitted us to obtain laboratory-quality images of living bacteria in the field. It might eventually be used as an *in situ* instrument to monitor High Arctic communities year around, once satellite systems are in place to be able to relay data.

Finally, we are working with NASA's Jet Propulsion Laboratory to package the miniaturized microscope into a commercial wet chemistry suite, the Robotic Chemistry Lab (RCAL) from Starsys Instruments, which is currently undergoing Mars Rover integration and desert testing. The instrument consists of a hermetically sealed, climate-controlled chamber similar to that of the Mars environmental compatibility assessment (MECA), which was scheduled to fly on the cancelled Mars 2001 Lander. Soil is collected by a scoop or drill and delivered to the chamber via a hopper. Issues of dye function at Martian pressures and temperatures are avoided within such a chamber, and dyes may be shipped lyophilized, increasing their stability.

Unlike MECA, which consisted of a single "teacup" or test tube into which a soil sample can be delivered, the robotic chemistry

lab has 20 individual test tubes into which water and electrolytes are placed for the journey to Mars. The test tubes are placed in a rotating carousel which allows them access to as many as four instruments, each connected to its own actuator. This first instrument is a stir/sonication probe which pierces the seal on the electrolyte reservoir in the test tube and delivers soil. Lyophilized dyes, singly or in combination, are placed in a sealed packet above the test tubes, so that piercing the seal with this probe causes their release into the solution. The ultraminiaturized microscope is approximately the size of one of the test tubes, and can be placed flush against it to focus automatically on any organisms or structures on the inner surface of the glass tube.

6. Future activities

6.1. Logistics

In 2005 the Canadian Space Agency began a formal program of analogue science through its Canadian Analogue Research Network (CARN) and its program of CARN grants to Canadian researchers to work at these and other analogue sites across Canada (Hipkin et al., 2007; Osinski et al., 2007). Along with the Haughton Mars Project on Devon Island and Pavillion Lake in British Columbia, the McGill Arctic Research Station was one of three sites funded under the program. M.A.R.S. obtained funding primarily to supply the technological and scientific infrastructure required for arctic field-based science in planetary exploration. This funding allows M.A.R.S. to provide the year-round technological management and science support and seasonal field logistical support needed for astrobiological and analogue studies at or near the McGill field facilities in the Canadian High Arctic.

The expansion of logistical facilities has effectively doubled the working capacity of the camp. The original McGill Station can support up to 10–12 scientists, and consists of a small cook hut, a research and accommodations building with both laboratory workspace and sleeping facilities, and three semi-permanent 'Weatherhaven' structures used for both storage and overflow accommodations. Power is provided to the station through a combination of gas powered generators and a small solar cell. Completed in 2007, the new CARN facility currently consists of one combined laboratory/kitchen building and one Weatherhaven for equipment storage, and is powered by a combination solar/wind system.

As no formal sleeping facilities are available at the CSA facility – accommodation is exclusively in tents – there is theoretically ample space for any number of researchers. It should be noted, though, that while the current expansion is indeed an ongoing process, we do not intend to increase the capacity of the camp significantly. However, we do intend to continue expanding both infrastructure (one to two new Weatherhavens for storage/emergency shelter) and communications (high-speed Ka band network).

6.2. Opportunities for collaboration

The astrobiology and analogue research activities that have been undertaken at M.A.R.S. are loosely organized under five space exploration themes (Table 2). The international nature of this research puts Canada, and M.A.R.S. in particular, in a favourable position with respect to analogue science and potential future collaborations.

Most of the research discussed in this paper is ongoing and will evolve as new opportunities arise. Currently, researchers at M.A.R.S. are looking to expand their teams to strengthen the Mars applications of five themes described. In the immediate future, evolving research topics will include: (i) testing of new exploration technologies (drills, geophysical tools, mobility systems); (ii) tele-medicine research; (iii) astronaut training, and; (iv) education and outreach activities. Future research is likely to be more mission-oriented and will involve the integration of several of the science activities around a central theme such as sample return or drilling. It is expected that direct partnerships with foreign programs and funding initiates (e.g. NASA's ASTEP program) will be developed in the near future.

At present, the primary limitation for collaboration is the size of potential research teams. We emphasize the outpost nature of our site by working in small manageable groups. In order to maintain this approach we expect to cap presence at 10–12 persons at each camp facility at any one time.

Potential collaborators interested in existing research must appreciate that current teams have priority and collaboration is seen as a way of expanding capacity, not duplicating the already productive science program. A second criterion is funding; because M.A.R.S. does not fund research, potential collaborators must already have funding sources in place and must be prepared to cover the costs of logistics and room and board.

In conclusion, interested parties have two methods by which to begin new research at M.A.R.S. Canadians interested in developing research should apply through CSA's CARN program, while foreign researchers should contact the M.A.R.S. Research Director directly.

Table 2
Analogue themes of research being conducted at M.A.R.S.

	Analogue themes	Projects and lead researchers
Theme 1	The search for water: perennial springs as analogues for hydrothermal systems on ancient Mars	P1. Low-temperature analogue of hydrothermal systems in thick permafrost (D. Andersen and W. Pollard) P2. Biomineralization under low temperature hydrothermal conditions (D. Andersen, H. Vali) P3. Geomorphology of perennial brine springs and freezing point depression. (W. Pollard, J. Heldmann and D. Andersen)
Theme 2	Water-related landforms: gully and valley formation	P4. Ice wedge polygons as analogues for Martian polygons (T. Haltigin and W. Pollard) P5. Ground ice stratigraphy (W. Pollard)
Theme 3	Drilling technology and sampling: Mars Autonomous Deep Drill (MADD)	P6. ASTID field testing Mars prototype (G. Briggs et al.–NASA Ames, JSC, Baker Hugues and McGill) P7. Contamination and Aseptic Sampling (L. Whyte, D. Juck, C. McKay, W. Pollard)
Theme 4	Life in extreme environments	P8. Cryptoendolithic communities in the Eureka area (C. Omelon and W. Pollard) P9. Life in ground ice (L. Whyte, B. Stevens, and W. Pollard) P10. Microbial ecology of perennial springs (L. Whyte, T. Neiderberger, D. Andersen)
Theme 5	Exploration: Mars Rover and Instrumentation Studies	P11. MDA: Mars rover based on Long Days Drive. P12. RIGID Capacitive-coupled resistivity techniques-OhmMapper (W. Pollard, T. Haltigin). P13. Mars GPR prototype field testing: (K. Williams, Buffalo State University) P. 14 Field microscopy (J. Nadeau)

7. Summary

Axel Heiberg Island provides a unique opportunity to examine the physical and biological processes characteristic of cold, dry environments. While research based out of the McGill Arctic Research Station has been ongoing for nearly 50 years, developments in the past decade have expanded to include investigations about the analogical value of this environment to propose hypotheses about similar processes on Mars. Presently, such analogue studies are continuing centered around geomorphic, hydrological, astrobiological, and technological research themes, and will be expanding greatly over the near-term. While, certainly, no single environment on Earth perfectly and completely replicates present-day Martian conditions, by using the Canadian High Arctic as a "natural laboratory" a wealth of information can be generated by furthering our understanding of polar desert systems and subsequently extrapolating ideas to the analogical target: Mars.

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References

Allan, C., Schiff, S., Peirson, D., English, M., Ecclestone, M., Adams, W.P., 1987. Colour Lake, Axel Heiberg Island, N.W.T., a naturall acid, high Arctic, lake—data report. In: Adams, P. (Ed.), Field Research on Axel Heiberg Island, N.W.T. McGill University, Canada.

Amundsen, H., Steele, A., Fogel, M., Mysen, B., Kiehle, J., Schweizer, M., Toporski, J., Treiman, A., 2004. Life in a Mars analogue: microbial activity associated with carbonate cemented lava breccia from NW Spitsbergen. In: 14th Annual V.M. Goldschmidt-Conference; Geochimica et Cosmochimica Acta 68 (11, Suppl. 1), A804.

Andersen, D.T., McKay, C.P., Wharton, R.A., Rummel, J.D., 1990. An Antarctic research outpost as a model for planetary exploration. Journal of the British Interplanetary Society 43, 499–504.

Andersen, D.T., McKay, C.P., Wharton, R.A., Rummel, J.D., 1992. Testing a Mars science outpost in the Antarctic Dry Valleys. Advances in Space Research 12 (5), 205–209.

Andersen, D.T., Pollard, W.H., McKay, C.P., Heldmann, J., 2002. Cold springs in permafrost on Earth and Mars. Journal of Geophysical Research 107, 1–7.

Andersen, D.T., Pollard, W.H., McKay, C.P., 2008. The perennial springs of Axel Heiberg Island as an analogue for groundwater discharge on Mars. In: Kane, D.L., Hinkel, K.M. (Eds.), Proceedings of the Ninth International Conference on Permafrost, University of Alaska Fairbanks, June 29–July 3, 2008, 43–48.

Arcone, S.A., Prentice, M.L., Delaney, A.J., 2002. Stratigraphic profiling with ground-penetrating radar in permafrost: a review of possible analogs for Mars. Journal of Geophysical Research 107 (E11), 5108.

Baker, V.R., 2001. Water and the Martian landscape. Nature 412, 228–236.

Barrett, J.E., Virginia, R.A., Wall, D.H., Cary, S.C., Adams, B.J., Hacker, A.L., Aislabie, J.M., 2006. Co-variation in soil biodiversity and biogeochemistry in Northern and Southern Victoria Land, Antarctica. Antarctic Science 18, 535–548.

Beschel, R.E., 1963. Sulfur springs at Gypsum Hill. Preliminary Report 1961–1962, Axel Heiberg Island Research Report, McGill University, pp. 183–187.

Blackhurst, R.L., Jarvis, K., Grady, M.M., 2004. Biologically-induced elemental variations in Antarctic sandstones: a potential test for Martian micro-organisms. International Journal of Astrobiology 3, 97–106.

Bowman, J.P., Nichols, D.S., 2005. Novel members of the family *Flavobacteriaceae* from Antarctic maritime habitats including *Subsaximicrobium wynnwilliamsii* gen. nov., sp. nov., *Subsaximicrobium saxinquilinus* sp. nov., *Subsaxibacter broadyi* gen. nov., sp. nov., *Lacinutrix copepodicola* gen. nov., sp. nov., and novel species of the genera *Bizionia*, *Gelidibacter* and *Gillisia*. International Journal of Systemic and Evolutionary Microbiology 55, 1471–1486.

Brasier, M.D., Green, O.R., Jephcoat, A.P., Kleppe, A.K., Van Krankendonk, M.J., Lindsay, J.F., Steele, A., Grassineau, N.V., 2007. Questioning the evidence for Earth's oldest fossils. Nature 416, 76–81.

Brass, G.W., 1980. Stability of brines on Mars. Icarus 42, 20–28.

Brinkmeyer, R., Knittel, K., Jürgens, J., Weyland, H., Amann, R., Helmke, E., 2003. Diversity and structure of bacterial communities in Arctic versus Antarctic pack ice. Applied and Environmental Microbiology 69, 6610–6619.

Bruchez Jr., M., Moronne, M., Gin, P., Weiss, S., Alivisatos, A.P., 1998. Semiconductor nanocrystals as fluorescent biological labels. Science 281, 2013–2016.

Cady, S.L., Farmer, J.D., Grotzinger, J.P., Schopf, J.W., Steele, A., 2003. Morphological biosignatures and the search for life on Mars. Astrobiology 3, 351–368.

Calvert, H.T., Dallimore, S.R., Hunter, J.A., 2001. Application of geophysical techniques for mapping ice-bearing sediments, Mackenzie Delta, Western Arctic, Canada. In: Proceedings of the conference on the geophysical Detection of Subsurface on Mars, #7041, Lunar and Planetary Institute, Houston, TX, USA, August 6–10, 2001.

Calvert, H.T., 2002. Capacitive-coupled resistivity survey of ice-bearing sediments, Mackenzie Delta, Canada. In: Society of Exploration Geophysicists. Proceedings of the 72nd annual meeting, Salt Lake City, UT, October. 6–11, 2002. Publisher: Society of Exploration Geophysicists, Tulsa, OK, USA.

Carpenter, E.L., Lin, S., Capone, D.G., 2000. Bacterial activity in South Pole snow. Applied and Environmental Microbiology 66, 4514–4517.

Christner, B.C., 2002. Incorporation of DNA and protein precursors into macromolecules by bacteria at −15 °C. Applied and Environmental Microbiology 68, 6435–6438.

Costard, F., Forget, F., Mangold, N., Peulvast, J.P., 2002. Formation of recent Martian debris flows by melting of near surface ground ice at high obliquity. Science 295, 110–113.

de la Torre, J.R., Goebel, B.M., Friedmann, E.I., Pace, N.R., 2003. Microbial diversity of cryptoendolithic communities from the McMurdo Dry Valleys, Antarctica. Applied and Environmental Microbiology 69, 3858–3867.

DePascale, G.P., Pollard, W.H., Williams, K.K., 2008. Geophysical mapping of ground ice using a combination of capacitive coupled resistivity and ground-penetrating radar, Northwest Territories, Canada. Journal of Geophysical Research 113, F02S90, doi:10.1029/2006JF000585.

Diggle, O., 2003. Statistical Analysis of Spatial Point Patterns. Arnold, London, UK.

Dutilleul, P., Haltigin, T.W., Pollard, W.H., 2009. Analysis of polygonal terrain landforms on Earth and Mars through spatial point patterns. Environmetrics 20 (2), 206–220.

Edlund, S.A., Alt, B.T., 1989. Regional congruence of vegetation and summer climate patterns in the Queen Elizabeth Islands, Northwest Territories, Canada. Arctic 42, 3–22.

Eicken, H., 2002. Studies of Earth Analogs of Potential Europan Habitats: Opportunities and Limitations. Europa Focus Group Workshop. University of Alaska, Fairbanks, pp. 11–12.

Fortier, D., Allard, M., 2004. Late Holocene syngenetic ice-wedge polygons development, Bylot Island, Canadian Arctic Archipelago. Canadian Journal of Earth Science 41, 997–1012 (Doi:10:1139.E04-031).

Friedmann, E.I., 1982. Endolithic microorganisms in the Antarctic cold desert. Science 215, 1045–1053.

Friedmann, E.I., 1986. The Antarctic cold desert and the search for traces of life on Mars. Advances in Space Research 6, 167–172.

Friedmann, E.I., Ocampo, R., 1976. Endolithic blue-green algae in the dry valleys: primary producers in the Antarctic desert ecosystem. Science 193, 1274–1279.

Friedmann, E.I., Ocampo-Friedmann, R., 1984. The Antarctic cryptoendolithic ecosystem: relevance to exobiology. Origins of Life 14, 771–776.

Friedmann, E.I., Friedmann, R.O., Weed, R., 1986. Trace fossils of endolithic microorganisms in Antarctica—a model for Mars. Origins of Life and Evolution of the Biosphere 16, 350.

Friedmann, E.I., McKay, C.P., Nienow, J.A., 1987. The cryptoendolithic microbial environment in the Ross Desert of Antarctica: satellite-transmitted continuous nanoclimate data, 1984–1986. Polar Biology 7, 273–287.

Friedmann, E.I., Weed, R., 1987. Microbial trace-fossil formation, biogenous, and abiotic weathering in the Antarctic Cold Desert. Science 236, 703–705.

Friedmann, E.I., Koriem, A., 1989. Life on Mars: how it disappeared (if it ever was there). Advances in Space Research 9, 167–172.

French, H.M., 1996. The Periglacial Environment. Longman Group, Essex, 341pp.

Gilichinsky, D., Rivkina, E., Bakermans, C., Shcherbakova, V., Petrovyskaya, L., Ozerskaya, S., Ivanushkina, N., Kochkina, G., Laurinavichuis, K., Pecheritsina, S., Fattakhova, R., Tieje, J.M., 2005. Biodiversity of cryopegs in permafrost. FEMS Microbiology Ecology 53, 117–128.

Goublic, S., Friedmann, I., Schneider, J., 1981. The lithobiontic ecological niche, with special reference to microorganisms. Journal of Sedimentary Petrology 51, 475–478.

Haberle, R.M., McKay, C.P., Schaeffer, J., Cabrol, N.A., Grin, E.A., Zent, A.P., Quinn, R.M., 2001. On the possibility of liquid water on present day Mars. Journal of Geophysical Research 106, 23,317–23,326.

Haldorsen, S., Heim, M., 1999. An Arctic groundwater system and its dependence upon climatic change: an example from Svalbard. Permafrost and Periglacial Processes 10, 137–149.
Haltigin, T.W., Osinski, G.R., Pollard, W.H., Dutilleul, P., 2007. Using spatial point patterns to quantify polar desert polygonal geometry. Lunar and Planetary Science XXXVIII, 1635.
Haltigin, T.W., Pollard, W.H., Osinski, G.R., Dutilleul, P., 2008a. Comparative morphometric analysis of polygonal terrain at potential Mars Phoenix landing sites. Lunar and Planetary Science XXXIX, 2475.
Haltigin, T.W., Pollard, W.H., Osinski, G.R., Dutilleul, P., Seaquist, J.W., 2008b. Spatial analysis of small-scale polygonal terrain in Utopia Planitia, Mars. In Kane, D.L., Hinkel, K.M. (eds.), Proceedings of the Ninth International Conference on Permafrost, University of Alaska Fairbanks, June 29–July 3, 2008, 639–644.
Heldmann, J.L., Mellon, M.T., 2004. Observations of Martian gullies and constraints on potential formation mechanisms. Icarus 156, 373–386.
Heldmann, J.L., Toon, O.B., Pollard, W.H., Mellon, M.T., Pitlick, J., McKay, C.P., Andersen, D.T., 2005. Formation of Martian gullies by the action of liquid water flowing under current Martian environmental conditions. Journal of Geophysical Research 110, E05004.
Hoën, E.W., 1964. The anhydrite diapirs of central western Axel Heiberg Island. Geology, No. 2, Axel Heiberg Island Research Reports, McGill University, Montreal, 102p.
Hipkin, V., Osinski, G.R., Berinstain, A., Leveille, R., 2007. The Canadian Analogue Research Network (CARN): opportunities for terrestrial analogue studies in Canada and Abroad. Lunar and Planetary Science XXXVIII, 2052.
Hinkel, K.M., Doolittle, J.A., Bockheim, J.G., Nelson, F.E., Paetzold, R., Kimble, J.M., Travis, R., 2001. Detection of subsurface permafrost features with ground-penetrating radar, Barrow, Alaska. Permafrost and Periglacial Processes 12, 179–190.
Hinrichs, K.U., Hayes, J.M., Sylva, S.P., Brewer, P.G., DeLong, E.F., 1999. Methane-consuming archaebacteria in marine sediments. Nature 398, 802–805.
Hubbard, G.S., Naderi, F.M., Garvin, J.B., 2002. Following the water, the new program for Mars exploration. Acta Astronautica 51 (1–9), 337–350.
Isaev, V.S., Abramenko, O.N., 2003. Demonstration of polygonal ice-wedges terrain on the Tazovsky Peninsula in the Western Siberia and some polygonal patterns on the Mars. Microsymposium on Comparative Planetology 38 (MS001), 1–2.
Jannasch, H.W., Mottl, M.J., 1985. Geomicrobiology of deep-sea hydrothermal vents. Science 229, 717–725.
Johnston, C.G., Vestal, J.R., 1989. Distribution of inorganic species in two Antarctic cryptoendolithic microbial communities. Geomicrobiology Journal 7, 137–153.
Juck, D.F., Whissell, G., Steven, B., Pollard, W., McKay, C.P., Greer, C.W., Whyte, L.G., 2005. Utilization of fluorescent microspheres and a green fluorescent protein-marked strain fro assessment of microbiological contamination of permafrost and ground ice core samples from the Canadian High Arctic. Applied and Environmental Microbiology 71 (2), 1035–1041.
Junge, K., Eicken, H., Deming, J.W., 2004. Bacterial activity at -2 to $-20\,^{\circ}C$ in Arctic wintertime sea ice. Applied and Environmental Microbiology 70, 550–557.
Kappen, L., 1993. Lichens in the Antarctic region. In: Friedman, E.I. (Ed.), Antarctic Microbiology. Wiley, New York.
Kawasaki, Y., 1999. Direct detection of Martian microorganisms based on fluorescence microscopy. Life Sciences: Exobiology 23, 309–317.
Kloepfer, J.A., Mielke, R.E., Wong, M.S., Nealson, K.H., Stucky, G., Nadeau, J., 2003. Quantum dots as strain- and metabolism-specific microbiological labels. Applied Environmental Microbiology 69, 4205–4213.
Kloepfer, J.A., Mielke, R.E., Nadeau, J.L., 2005. Uptake of CdSe and CdSe/Zns quantum dots into bacteria via purine-dependent mechanisms. Applied and Environmental Microbiology 71, 2548–2557.
Lachenbruch, A.H., 1962. Mechanics of thermal contraction cracks and ice-wedge polygons in permafrost. Special paper to the Geological Society of America 70, 69pp.
Lauritzen, S., Bottrell, S., 1994. Microbiological activity in thermoglacial karst springs, south Spitsbergen. Geomicrobiology 12, 161–173.
Mackay, J.R., 1993. Air temperature, snow cover, creep of frozen ground, and the time of ice-wedge cracking, western Arctic Coast. Canadian Journal of Earth Sciences 30, 1720–1729.
Mackay, J.R., 1999. Periglacial features developed on the exposed lake bottoms of seven lakes that drained rapidly after 1950, Tuktoyaktuk Peninsula area, Western Arctic coast, Canada. Permafrost and Periglacial Processes 10, 39–63.
Malin, C.M., Edgett, K.S., Posiolova, L.V., McColley, S.M., Noe Dobrea, E.Z., 2006. Present-day impact catering rate and contemporary gully activity on Mars. Science 314, 1573–1577.
Mangold, N., 2005. High latitude patterned grounds on Mars: classification, distribution and climatic control. Icarus 174, 336–359.
Mangold, N., Maurice, S., Feldman, W.C., Costard, F., Forget, F., 2004. Spatial relationships between patterned ground and ground ice detected by the Neutron Spectrometer on Mars. Journal of Geophysical Research 109 (E08001) (Doi:10.129/2004JE002235).
Marion, G.M., 1997. A theoretical evaluation of mineral stability in Don Juan Pond, Wright Valley, Victoria Land. Antarctic Science 9, 92–97.
McEwen, A.S., Hansen, C.J., Delamere, W.A., Eliason, E.M., Herkenhoff, K.E., Keszthelyi, L., Gulick, V.C., Kirk, R.L., Mellon, M.T., Grant, J.A., Thomas, N., Weitz, C.M., Squyres, S.W., Bridges, N.T., Murchie, S.L., Seelos, F., Seelos, K., Okubo, C.H., Milazzo, M.P., Tornabene, L.L., Jaeger, W.L., Byrne, S., Russell, P.S., Griffes, J.L., Martínez-Alonso, S., Davatzes, A., Chuang, F.C., Thomson, B.J., Fishbaugh, K.E., Dundas, C.M., Kolb, K.J., Banks, M.E., Wray, J.J., 2007. A closer look at water-related geological activity on Mars. Science 317, 1706–1709.
McKay, C.P., Friedmann, E.I., 1985. The cryptoendolithic microbial environment in the Antarctic cold desert: temperature variations in nature. Polar Biology 4, 19–25.
McKay, C.P., Andersen, D.T., Pollard, W.H., Heldmann, J.L., Doran, P.T., Fritsen, C.H., Priscu, J.C., 2005. Polar lakes, streams, and springs as analogs for the hydrological cycle on Mars. In: Tokana, T. (Ed.), Water on Mars and Life. Springer, Berlin.
Mikucki, J.A., Priscu, J.C., 2007. Bacterial diversity associated with Blood Falls, a subglacial outflow from the Taylor Glacier, Antarctica. Applied and Environmental Microbiology 73, 4029–4039.
Milsom, J., 2003. Field Geophysics, 3rd ed. Wiley, Chichester, 232pp.
Muller, F., 1963. Investigations in an ice shaft in the accumulation zone of the McGill Ice Cap. Preliminary report 1961–1962, Axel Heiberg Island Research Report. McGill University, Montreal, Quebec, pp. 27–36.
Nadeau, J.L., Perreault, N.N., Niederberger, T.D., Whyte, L.G., Sun, H.J., Leon, R., 2008. Fluorescence microscopy as a tool for *in situ* life detection. Astrobiology 8 (4), 859–874.
Nedashkovskaya, O.I., Kim, S.B., Lee, K.H., Mikhailov, V.V., Bae, K.S., 2005. *Gillisia mitskevichiae* sp. nov., a novel bacterium of the family Flavobacteriaceae, isolated from sea water. International Journal of Systemic and Evolutionary Microbiology 55, 321–323.
Niederberger, T.D., Perreault, N.N., Nadeau, J., Mielke, R.E., Greer, C.W., Andersen, D.T., Whyte, L.G., 2008. Filamentous sulfur-oxidizing communities thriving in a unique cryo-environment of the Canadian high Arctic. Applied and Environmental Microbiology (in review).
Omelon, C.R., 2008. Endolithic microbial communities in polar desert habitats. Geomicrobiology Journal 25.
Omelon, C.R., Pollard, W.H., Marion, G.M., 2001. Seasonal formation of ikaite ($CaCO_3*6H_2O$) in saline spring discharge at Expedition Fiord, Canadian high Arctic: assessing conditional constraints for natural crystal growth. Geochimica et Cosmochimica Acta 65 (9), 1429–1437.
Omelon, C.R., Pollard, W.H., Andersen, D.T., 2006a. A geochemical evaluation of perennial spring activity and associated mineral precipitates at Expedition Fjord, Axel Heiberg Island, Canadian High Arctic. Applied Geochemistry 21, 1–15.
Omelon, C.R., Pollard, W.H., Ferris, F.G., 2006b. Environmental controls on microbial colonization of high Arctic cryptoendolithic habitats. Polar Biology 30, 19–29.
Omelon, C.R., Pollard, W.H., Ferris, F.G., 2006c. Chemical and ultrastructural characterization of high Arctic cryptoendolithic habitats. Geomicrobiology Journal 23, 189–200.
Omelon, C.R., Pollard, W.H., Ferris, F.G., 2007. Inorganic species distribution and microbial diversity within high Arctic cryptoendolithic habitats. Microbial Ecology 54, 740–752.
Omelon, C.R., Pollard, W.H., Ferris, F.G., Bennett, P.C., 2008. Cyanobacteria within cryptoendolithic habitats: the role of high pH in biogenic rock weathering in the Canadian high Arctic. In: Kane, D.L., Hinkel, K.M. (Eds.), Proceedings of the Ninth International Conference on Permafrost, University of Alaska Fairbanks, June 29–July 3, 2008, 1327–1332.
Onstott, T.C., McGown, D., Kessler, J., Sherwood Lollar, B., Lehmann, K.K., Clifford, S.M., 2006. Martian CH_4: sources, flux and detection. Astrobiology 6, 377–395.
Osinski, G.R., Leveille, R., Berinstain, A., Lebeuf, M., Bamsey, M., 2007. Terrestrial analogues to Mars and the Moon: Canada's role. Geoscience Canada 33 (4), 175–188.
Pelletier, J.D., Kolb, K.J., McEwen, A.S., Kirk, R.L., 2008. Recent bright gully deposits on Mars; wet or dry flow? Geology (Boulder) 36 (3), 211–214.
Perreault, N.N., Andersen, D.T., Pollard, W.H., Greer, C.W., Whyte, L.G., 2007. Characterization of the prokaryotic diversity in cold saline perennial springs in the Canadian high Arctic. Applied and Environmental Microbiology 73, 1532–1543.
Perreault, N.N., Greer, C.W., Andersen, D.T., Tille, S., Lacrampe-Couloume, G., Sherwood Lollar, B., Whyte, L.G., 2008. Heterotophic and autotrophic microbial populations in cold perennial springs of the high Arctic. Applied and Environmental Microbiology (in review).
Pollard, W.H., 1991. A high Arctic occurrence of seasonal frost mounds. In: Proceedings of the Northern Hydrology Symposium, Saskatoon, SK, pp. 263–275.
Pollard, W., Bell, T., 1998. Massive ice formation in the Eureka Sound Lowlands: A landscape model. In: Lewkowicz, A.G., Allard, M. (Eds.), Proceedings of the Seventh International Permafrost Conference, Yellowknife, 23 27 June, Université Laval, Centre d'etudes nordiques, Collection Nordicana, No 57, pp. 903–908.
Pollard, W.H., Omelon, C., Andersen, D.T., McKay, C.P., 1999. Perennial spring occurrence in the expedition fiord area of western Axel Heiberg Island, Canadian high Arctic. Canadian Journal of Earth Sciences 36, 105–120.
Rivkina, E.M., Friedmann, E.I., McKay, C.P., Gilichinsky, D.A., 2000. Metabolic activity of permafrost bacteria below the freezing point. Applied and Environmental Microbiology 66, 3230–3233.
Rogers, J.D., Karkkainen, A.H.O., Tkaczyk, T., Rantala, J.T., Descour, M.R., 2004. Realization of refractive microoptics through grayscale lithographic patterning of photosensitive hybrid glass. Optics Express 12, 1294–1303.
Rossbacher, L.A., 1986. Nearest-neighbour analysis: a technique for quantitative evaluation of polygonal ground patterns. Geografiska Annaler, Series A, Physical Geography 68 (1/2), 101–105.
Schiff, S., English, M., Ecclestone, M., Elgood, R., Hinton, M., Pezzutto, L., 1991. Constraints on the origin of acidity in Colour Lake, Axel Heiberg Island. In:

Proceedings of the Northern Hydrology Symposium, Saskatoon, SK, pp. 303–318.
Seibert, N.M., Kargel, J.S., 2001. Small-scale Martian polygonal terrain: implications for liquid surface water. Geophysical Research Letters 28 (5), 899–902.
Sletten, R.S., Hallet, B., Fletcher, R.C., 2003. Resurfacing time of terrestrial surfaces by the formation and maturation of polygonal patterned ground. Journal of Geophysical Research 108 (E4), 8044.
Stackebrandt, E., Brambilla, E., Cousin, S., Dirks, W., Pukall, R., 2004. Culture-independent analysis of bacterial species from an anaerobic mat from Lake Fryxell, Antarctic: prokaryotic diversity revisited. Cellular and Molecular Biology 50, 517–524.
Stephenson, R., Van Berkel, J., Cloethingh, S., 1992. Relation between salt diapirism and tectonic history of the Sverdrup Basin, Arctic Canada. Canadian Journal of Earth Sciences 29, 2640–2652.
Steele, A., Schweizer, M., Amundsen, H.E.F., Wainwright, N., 2004. In-field testing of life detection instruments and protocols in a Mars analogue arctic environment. AbSciCon 2004, NASA Ames Research Center, Moffett Field, CA; International Journal of Astrobiology Supplement 1 (March), 24.
Steven, B., Briggs, G., McKay, C.P., Pollard, W.H., Greer, C.W., Whyte, L.G., 2007a. Characterization of the microbial diversity in a permafrost sample from the Canadian high Arctic using culture-dependent and culture-independent methods. FEMS Microbiology Ecology 59, 513–523.
Steven, B., Niederberger, T.D., Bottos, E.M., Dyen, M.R., Whyte, L.G., 2007b. Development of a sensitive radiorespiration method for detecting microbial activity at subzero temperatures. Journal of Microbiological Methods 71, 275–280.
Sun, H.J., Friedmann, E.I., 1999. Growth on geological time scales in the Antarctic cryptoendolithic microbial community. Geomicrobiology Journal 16, 193–202.
Taylor, A.E., Judge, A.S., 1976. Canadian Geothermal Data Collection—Northern Wells 1975, Earth Physics Branch, Energy, Mines and Resources, Canada, Ottawa, Canada, 142pp.
Thorsteinsson, R., Tozier, E.T., 1970. Geology of the Arctic Archipelago. Geology and EconomicMminerals of Canada. Part B. In: Douglas, R. (Ed.), Geological Survey of Canada, Economic Geology Report No. 1, pp. 548–590.
Vali, H., Pollard, W., Andersen, D., Vali, H., McKay, C., Arkani-Hamed, J., 1999. A Mars Analogue Research Station (M.A.R.S.) in the Canadian High Arctic. In: Proceedings Canada in Space Exploration: The Second Canadian Space Exploration Workshop, Canada Space Agency, University of Calgary, Calgary, Alberta, October 17–21. pp. 31–40.
van Everdingen, R.O., 2002. Multi-language glossary of permafrost and related ground0ice terms, Boulder, CO: National Snow and Ice Data Center/World Data Center for Gloaciology, 200.
Van Trappen, S., Vandecandelaera, I., Mergaert, J., Swings, J., 2004a. *Gillisia limnaea* gen. nov., sp. nov., a new member of the family *Flavobacteriaceae* isolated from a microbial mat in Lake Fryxell, Antarctica. International Journal of Systemic and Evolutionary Microbiology 54, 445–448.
Van Trappen, S., Mergaert, J., Swings, J., 2004b. *Loktanella salsilacus* gen. nov., sp. nov., *Loktanella fryxellensis* sp. nov and *Loktanella vestfoldensis* sp. nov., new members of the *Rhodobacter* group microbial mats in Antarctic lakes. International Journal of Systemic and Evolutionary Microbiology 54, 1263–1269.
Wagner, D., Gattinger, A., Embacher, A., Pfeiffer, E.M., Schloter, M., Lipski, A., 2007. Methanogenic activity and biomass in Holocene permafrost deposits of the Lena Delta, Siberian Arctic and its implication for the global methane budget. Global Change Biology 13, 1089–1099.
Walker, J.J., Pace, N.R., 2007. Phylogenetic composition of rocky mountain endolithic microbial ecosystems. Applied and Environmental Microbiology 73, 3497–3504.
Wierzchos, J., Ascaso, C., 2002. Microbial fossil record of rocks from the Ross Desert, Antarctica: implications in the search for past life on Mars. International Journal of Astrobiology 1, 51–59.
Wierzchos, J., Ascaso, C., Sancho, L.G., Green, A., 2003. Iron-rich diagenetic minerals are biomarkers of microbial activity in Antarctic rocks. Geomicrobiology Journal 20, 15–24.
Wierzchos, J., Sancho, L.G., Ascaso, C., 2005. Biomineralization of endolithic microbes in rocks from the McMurdo Dry Valleys of Antarctica: implications for microbial fossil formation and their detection. Environmental Microbiology 7, 566–575.
Williams, J.R., van Everdingen, R.O., 1973. Groundwater investigations in permafrost regions of North America: a review. In: Permafrost, Second International Conference, North American Contribution, National Academy of Sciences, Washington, DC, pp. 435–446.
Williams, K.K., Grant, J.A., Schutz, A.E., 2005. Ground-penetrating radar in Mars analog terrains; testing the *Strata* instrument. Workshop on Radar Investigations, d-6019.
Worsley, P., Gurney, S.D., 1996. Geomorphology and hydrogeological significance of the Holocene pingos in the Karup Valley, Traill Island, northern east Greenland. Journal of Quaternary Science 11, 249–262.
Wynn-Williams, D.D., Edwards, H.G.M., 2000a. Antarctic ecosystems as models for extraterrestrial surface habitats. Planetary and Space Science 48, 1065–1075.
Wynn-Williams, D.D., Edwards, H.G.M., 2000b. Proximal analysis of regolith habitats and protective biomolecules in situ by laser Raman spectroscopy: overview of terrestrial Antarctic habitats and Mars analogs. Icarus 144, 486–503.